Weekly Progress Report

Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 8 – September 14, 2025

What was accomplished last week:

- Finalized preliminary proposal including problem statement, objectives, and methodology
- Completed hardware requirements analysis and budget estimation (~$5,400)
- Researched regulatory compliance requirements (HIPAA, NIST CSF, IEC 62443, ISO/IEC 27001)
- Ordered Dell PowerEdge R740XD server and storage components
- Ordered UniFi networking equipment (UDM-SE, Switch Pro Max 24 PoE, U6 Pro AP)
- Began literature review on hospital cybersecurity threats and IoT vulnerabilities

Plans for coming week:

- Await hardware delivery (expected mid-week)
- Prepare rack space and power infrastructure
- Download TrueNAS CORE ISO and documentation
- Design initial VLAN architecture (IoT, Management, Secure, Guest)
- Order remaining Raspberry Pi 5 kits and SenseHAT components

Issues or concerns:

- Lead time on some hardware components may extend into next week
- Need to finalize rack mounting strategy for optimal cable management

Advisor comment(s):

Signature of Advisor/Date: _____

Weekly Progress Report

Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 15 – September 21, 2025

What was accomplished last week:

- Received Dell PowerEdge R740XD server and Sysracks 18U rack
- Received UniFi UDM-SE gateway and Switch Pro Max 24 PoE
- Physically racked server using Dell 2U B6 sliding readyrails
- Installed CyberPower CP1500 UPS and connected to rack PDU
- Performed initial power-on test of server - verified POST and iDRAC9 functionality
- Connected UniFi equipment for initial configuration
- Received 5x Raspberry Pi 5 kits with PoE+ HATs and SenseHAT displays

Plans for coming week:

- Install TrueNAS CORE on Dell R740XD server
- Configure iDRAC9 for remote management
- Set up initial network configuration with UDM-SE
- Begin VLAN segmentation planning (VLAN 10: IoT, VLAN 20: Medical, VLAN 30: Management, VLAN 40: Guest)
- Configure ZFS storage pools on TrueNAS

Issues or concerns:

- Need to decide on RAID configuration for optimal performance vs redundancy
- iDRAC9 default credentials need to be changed for security

Advisor comment(s):

Signature of Advisor/Date: _____

Weekly Progress Report

Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 22 – September 28, 2025

What was accomplished last week:

• Successfully installed TrueNAS CORE on Dell R740XD using BOSS 2x 240GB M.2 SSDs (RAID 1)
• Configured iDRAC9 with secure credentials and verified remote access capability
• Created ZFS storage pools: RAID 1 on 2x 480GB SSDs (~480GB usable) for VMs
• Set up initial network interface configuration on TrueNAS (X550 10GbE primary, I350 1GbE backup)
• Configured UDM-SE with production-style segmented VLAN architecture:
  * VLAN 10 (10.10.0.0/24): IoT/Patient Sensors
  * VLAN 20 (10.20.0.0/24): Medical Services
  * VLAN 30 (10.30.0.0/24): Secure Management
  * VLAN 40 (10.40.0.0/24): Guest Network
• Connected Switch Pro Max 24 PoE to UDM-SE and verified 10GbE uplink using SFP+ DAC cable

Plans for coming week:

• Deploy first VMs: OPNsense firewall and MQTT broker
• Configure switch port profiles for VLAN assignments
• Begin Raspberry Pi 5 initial setup and OS installation
• Test PoE functionality with Raspberry Pi devices
• Start developing Python scripts for SenseHAT patient vital simulation

Issues or concerns:

• Complex inter-VLAN routing rules need careful planning to avoid misconfigurations
• Need to ensure proper firewall rules between medical services and management networks

• Implementing production-level network segmentation from the start - will monitor complexity

Advisor comment(s):

Signature of Advisor/Date: _____

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Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 29 – October 5, 2025

What was accomplished last week:

- Deployed OPNsense Firewall VM on TrueNAS (2 vCPU, 4GB RAM, multiple VLAN bridges)
- Configured switch port assignments for VLAN segregation
- Set up ports 2, 3, 4 on switch for VLAN 10 (IoT devices)
- Installed Raspberry Pi OS on all 5 Raspberry Pi 5 units
- Successfully tested PoE+ functionality - Pis powered and networked via single cable
- Attached SenseHAT displays to 3 Raspberry Pis
- Installed required Python libraries (sense-hat, paho-mqtt) on Raspberry Pis
- Configured static IP addressing for Pis on VLAN 10 (10.10.0.20, .21, .22)

Plans for coming week:

- Deploy MQTT broker VM (Mosquitto) for IoT communications
- Develop ECG simulation Python scripts for realistic heartbeat displays
- Create temperature monitoring scripts with encryption (HIPAA compliance)
- Test MQTT publish/subscribe between Raspberry Pis and broker
- Begin documentation of network topology

Issues or concerns:

- Raspberry Pi SenseHAT temperature readings affected by CPU heat - need calibration
- VLAN routing complexity increasing - firewall rules between VLANs proving challenging
- Inter-VLAN communication requiring more troubleshooting than anticipated

Advisor comment(s):

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Weekly Progress Report

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Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 6 – October 12, 2025

What was accomplished last week:

• Deployed MQTT Broker VM (Ubuntu 24.04, Mosquitto) on TrueNAS - 2 vCPU, 4GB RAM
• Configured Mosquitto with TLS encryption for secure communications
• Developed HeartRate1Monitor.py - ECG-style heartbeat sweep display using Gaussian waveform modeling
• Developed HeartRate2Monitor.py - Secondary ECG simulator with different patient ID
• Successfully tested ECG scripts on Raspberry Pi with SenseHAT - 62 BPM realistic waveform
• Created temp_scroll.py - basic temperature display script
• Developed secure_temp_logger.py - HIPAA-compliant encrypted temperature logging using Fernet
• Generated encryption key and stored securely: /home/r-pi-1/.keys/patient_zero.key

Plans for coming week:

• Deploy Node-RED VM for IoT workflow automation
• Test end-to-end MQTT data flow: Pi sensors → MQTT broker → Node-RED
• Configure systemd services for persistent ECG monitoring across reboots
• Begin PostgreSQL database VM deployment for data storage
• Create initial Grafana dashboard mockup for vital signs visualization

Issues or concerns:

• Attempted cloud storage integration (OneDrive/rclone) - proved problematic and unreliable
• Decided to use local file storage instead for better reliability
• VLAN segmentation causing significant connectivity issues between IoT devices and VMs
• Need to document encryption key management procedures for compliance

Advisor comment(s):

Signature of Advisor/Date: _____

Weekly Progress Report

Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 13 – October 19, 2025

What was accomplished last week:

• Deployed Node-RED VM (Ubuntu, 2 vCPU, 4GB RAM) on VLAN 20
• Configured Node-RED flows for MQTT data ingestion and processing
• Attempted MQTT publish from Raspberry Pi to broker - encountering inter-VLAN routing issues
• Configured systemd service files for HeartRate1Monitor.py and HeartRate2Monitor.py
• Verified ECG monitoring scripts persist across system reboots
• Set up ID1.py script for patient identification display with blinking effect
• Tested encrypted temperature logging - verified 30-second intervals and Fernet encryption
• Successfully decrypted temperature logs using decrypt_log.py

Plans for coming week:

• Troubleshoot and resolve VLAN connectivity issues
• Integrate MQTT data flow with Node-RED processing
• Begin PostgreSQL VM deployment
• Create database schema for storing patient vital signs
• Update Python scripts to publish sensor data to MQTT topics

Issues or concerns:

• CRITICAL: Multi-VLAN architecture creating severe complexity in data flows
• Firewall rules between VLANs requiring extensive troubleshooting - preventing proper IoT integration
• Raspberry Pis on VLAN 10 cannot reliably communicate with MQTT broker on VLAN 20
• LESSON LEARNED: Jumped directly to production-type segmented network architecture too early

• May need to simplify network architecture to prioritize functionality before security segmentation

Advisor comment(s):

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Weekly Progress Report

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Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 20 – October 26, 2025

What was accomplished last week:

• Attempted MQTT broker and Node-RED integration - unsuccessful due to VLAN issues
• Verified TLS 1.3 encryption configuration on MQTT (when accessible)
• Documented current VM infrastructure: mqtt_broker_vm and node_red_vm deployed but not fully functional
• Created network topology diagram showing complex VLAN segmentation
• Extensively tested inter-VLAN routing rules via OPNsense firewall - persistent connectivity problems
• Researched best practices for phased network deployment in lab environments
• Made critical project decision: flatten network architecture temporarily
• Backed up TrueNAS configuration to laptop for disaster recovery before making changes

Plans for coming week:

• Redesign network architecture - flatten VLANs to single subnet (VLAN30: 10.30.0.0/24) for initial build
• Create NETWORK_SETUP_PROGRESS.md document to track configuration changes and lessons learned
• Execute migration from segmented multi-VLAN to flat single-subnet architecture
• Prepare for network reconfiguration while preserving VM functionality
• Document decision rationale and plan for future VLAN reinstatement

Issues or concerns:

• MAJOR LESSON LEARNED: Attempting production-level security architecture at project start was a mistake
• Multi-VLAN segmentation preventing completion of core IoT functionality
• Raspberry Pis losing connectivity, data flows blocked by complex firewall rules

• DECISION: Prioritize building functional infrastructure first, add security layers incrementally later
• Will reinstate multi-level segmented VLANs once core data pipeline is stable and tested
• This reflects real-world principle: establish baseline functionality before implementing complex security

Advisor comment(s):

Signature of Advisor/Date: _____

Weekly Progress Report

Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 27 – November 2, 2025

What was accomplished last week:

• Created comprehensive NETWORK_SETUP_PROGRESS.md documenting architectural pivot
• Backed up TrueNAS configuration (restoration file saved to laptop)
• Successfully flattened network to single subnet (VLAN30: 10.30.0.0/24):
  * UDM-SE: 10.30.0.1 (gateway)
  * TrueNAS: 10.30.0.44/24 on br0 interface
  * DHCP range: 10.30.0.6 - 10.30.0.254
• Removed test IP configurations from TrueNAS (cleaned up 192.168.1.10)
• Reconfigured switch ports 2, 3, 4 from VLAN10 to VLAN30 - all devices now on same subnet
• Power cycled Raspberry Pis via PoE to force DHCP renewal on new network
• Verified 2 Pis still running ECG scripts (HeartRate1Monitor and HeartRate2Monitor)
• Confirmed mqtt_broker_vm and node_red_vm operational after network simplification

Plans for coming week:

• Manually check each Raspberry Pi to find new IP addresses (10.30.0.x range)
• SSH to each Pi and verify connectivity to TrueNAS (10.30.0.44) and MQTT broker
• Configure static IP addresses for Pis (recommend 10.30.0.20, .21, .22, .23, .24)
• Test ping between all devices on new flat network - expect immediate success
• Delete OPNsense VM from TrueNAS (no longer needed with simplified architecture)
• Document plan for eventual VLAN reinstatement after core functionality proven

Issues or concerns:

• Raspberry Pis may have cached old network configurations - need manual verification
• Unknown if all Pis successfully obtained new DHCP addresses on 10.30.0.x subnet

• Need systematic approach to discover Pi IPs without causing disruption
• NOTE: This flat network is TEMPORARY - will reinstate security segmentation in future phase

Advisor comment(s):

Signature of Advisor/Date: _____

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Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: November 3 – November 9, 2025

What was accomplished last week:

• Successfully located all Raspberry Pi IP addresses on new 10.30.0.0/24 network
• Verified connectivity via SSH to operational Raspberry Pis - IMMEDIATE SUCCESS
• Confirmed HeartRate1Monitor.py and HeartRate2Monitor.py scripts still running perfectly
• Tested network connectivity: Pis can now reach TrueNAS (10.30.0.44) and UDM-SE (10.30.0.1)
• BREAKTHROUGH: MQTT broker VM now accessible from Raspberry Pis - no firewall blocking
• Deleted OPNsense VM from TrueNAS - unnecessary with flat network architecture
• Updated network documentation with current simplified topology
• Tested "portable mode" - disconnected WAN cable, verified lab functions standalone
• Network simplification vindicated - all connectivity issues resolved

Plans for coming week:

• Configure static IP addresses for all Raspberry Pis (10.30.0.20-24)
• Update Raspberry Pi scripts to publish data to MQTT broker on 10.30.0.51
• Deploy PostgreSQL database VM (10.30.0.50 recommended)
• Deploy Grafana/Prometheus VM (10.30.0.52 recommended)
• Create MQTT → PostgreSQL data pipeline (now possible with flat network)
• Begin building Grafana dashboards for real-time vital signs visualization

Issues or concerns:

• Need to update all Python scripts with new MQTT broker IP address
• Static IP assignment will require editing /etc/dhcpcd.conf on each Pi
• Should document IP allocation scheme for future VM deployments (10.30.0.50+)

• Must plan timeline for reinstating multi-level VLAN segmentation once core infrastructure proven

Advisor comment(s):

Signature of Advisor/Date: _____

Weekly Progress Report

Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: November 10 – November 16, 2025

What was accomplished last week:

• Assigned static IP addresses to operational Raspberry Pis (10.30.0.20, .21)
• Updated MQTT broker configuration to accept connections on 10.30.0.51
• Successfully verified end-to-end connectivity: Raspberry Pis → MQTT Broker → Node-RED
• Documented IP allocation scheme:
  * 10.30.0.1: UDM-SE Gateway
  * 10.30.0.44: TrueNAS
  * 10.30.0.20-24: Raspberry Pi IoT sensors
  * 10.30.0.50+: Future VM infrastructure (PostgreSQL, Grafana, FHIR, PACS, Wazuh, etc.)
• Created VM deployment priority list organized in four tiers
• Prepared 5-minute capstone presentation demonstrating project scope and current progress
• Documented encryption key: 732a7722df86ee181382dfc13a5fb02041a09f341174283a9e0d4f4ef79ae9c4
• Reflected on architectural decision: Lesson learned about phased implementation approach

Plans for coming week:

• Deploy PostgreSQL database VM for storing patient vital signs
• Configure database schema for ECG data, temperature logs, and timestamps
• Update Python scripts to publish structured JSON data via MQTT
• Create Node-RED flow to insert MQTT data into PostgreSQL
• Begin Grafana VM deployment for dashboard visualization
• Prepare demonstration of current ECG monitoring functionality
• Plan future phase: Reinstate multi-level segmented VLANs after core data pipeline proven stable

Issues or concerns:

- Need to decide on database schema design for optimal query performance
- Consider time-series database (TimescaleDB) vs traditional PostgreSQL
- Remaining 3 Raspberry Pis need configuration and script deployment
- AI camera Pi is lower priority - focus on vital signs monitoring first
- Future work: Document plan for phased VLAN segmentation reinstatement
- Key takeaway: Building complex security architecture requires functional foundation first

Advisor comment(s):

Signature of Advisor/Date: _____